SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON: A STATEMENTS-AND-FIGURES READING EDITION

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Abstract. This reading edition records the definitions, formal statements, explanatory prose, and figures of the complete paper while omitting every proof body and the calculation details used to establish the results. Its main statement is that a regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The canonical C-triangle/V-triangle classification routes a hypothetical cover to four mechanisms: trace-length deficit, propagation of boundary-reach lower bounds, area loss, and an equilateral-enclosure obstruction for nine forced points. The unabridged paper and exact certificates remain the authority for all proofs.

Scope of this edition. All theorem, lemma, proposition, corollary, definition, and remark statements are retained, together with the paper's explanatory prose and figures. Proof environments and calculation-only certificate material are omitted. The unabridged paper and exact certificates remain the proof authority.

Contents

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1. Introduction and Result Outline

1.1. The theorem and the canonical labeling. Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \qquad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

For a hypothetical cover, use the canonical labeling

$$U_C, U_0, \dots, U_5$$

with

$$O \in U_C, \qquad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at V_i*. Set

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

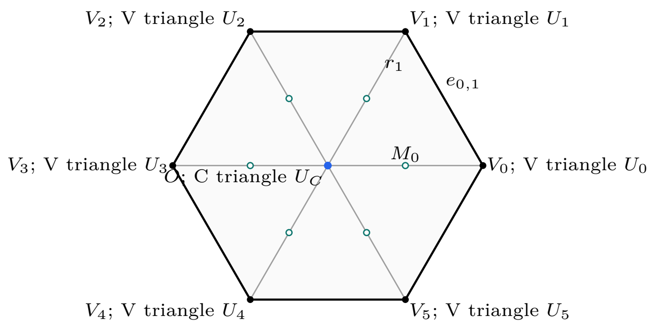


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H.

1.2. **Finite classifications.** The center type is the number of boundary edges on which T_C has a positive-length trace. The only possibilities are $0, 1, 2$, denoted by $\mathrm{CE0}, \mathrm{CE1}, \mathrm{CE2}$.

For the V triangle T_i, let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|,$$

with indices modulo 6. The exhaustive types are

type	$(o(T_i), n(T_i))$
Vd0	$(1,0), (2,0), (3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Let A_i, B_i, C_i be the maximal reaches of T_i toward V_{i-1}, V_{i+1}, O, respectively. The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds for these reaches and never the maxima defining N_+.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1}, the two incident open traces have the form $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a *boundary gap*; equality gives a singleton gap, which remains uncovered. Let N_{gap} be the number of positive center traces that contain a boundary gap.

Let N_{sp} be the number of V triangles of type $\mathrm{Vd1}, \mathrm{Vd2}$, or T3-like. The corresponding count of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\mathrm{sp}}.$$

Every CE1/CE2 configuration with $N_+ + N_{\mathrm{sp}} \geq 3$ is excluded by the trace-length statement on S.

1.3. **Exhaustive routing.** Table 1 is read from top to bottom in the CE1/CE2 part. The first applicable row is the unique method for a hypothetical cover. The full The full classification is Proposition 2.13.

center type and N_+	exhaustive refinement	method
CE0, $N_+ = 0$	all vertex types	1
CE0, $N_+ = 1$	all six triangles Vd0	4
CE0, $N_+ = 1$	a Vd1 or Vd2 triangle	1
CE0, $N_+ = 1$	no Vd1/Vd2 triangle and a T3-like triangle	3
CE0, $N_+ \geq 2$	all vertex types	3
CE1/CE2	$N_+ + N_{\mathrm{sp}} \geq 3$	1
CE1/CE2, $N_+ = 0$	all six triangles Vd0	2
CE1/CE2, $N_+ = 0$	a Vd1 or Vd2 triangle	1
CE1/CE2, $N_+ = 0$	no Vd1/Vd2 triangle and a T3-like triangle	2
CE1/CE2, $N_+ = 1$	all Vd0, no boundary gap	4
CE1/CE2, $N_+ = 1$	all Vd0, a boundary gap	2
CE1/CE2, $N_+ = 1$	exactly one T3-like, no Vd1/Vd2	2
CE1/CE2, $N_+ = 1$	exactly one Vd1/Vd2, no T3-like	1 / 1+2

Table 1. The exhaustive routing. Method 1 is a trace-length argument, Method 2 propagates reach bounds, Method 3 sums area loss, and Method 4 uses a nine-point enclosure obstruction. In the last row, the first alternative is for CE1 and the second for CE2.

1.4. **Organization of the reading edition.** Section 2 records the structural classification, strict handoff selection, midpoint facts, and signed C-triangle normal form. Sections 3–6 record the statements for the four mechanisms, and Section 7 records the exhaustive conclusion. Proof bodies and calculation details are omitted throughout.

2. Structural Reduction and Common Geometry

This section records the finite reduction used by every later method: the open–closed scaling equivalence, the C-triangle and V-triangle classifications, strict boundary handoffs, midpoint facts, exhaustive routing, and the signed center normal form used by the trace and propagation arguments. The proofs are omitted in this reading edition.

2.1. **Structural Reduction to Finite Cases.** Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{1}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L. All symmetries used below belong to the dihedral group D_6; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1.1. *Open, closed, and scaled formulations.*

Proposition 2.1 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *The set H is covered by seven open equilateral triangles of side length* 1*.*
(2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
(3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length* 1*.*

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

and these seven triangles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

2.1.2. *The center classification.* For a closed C triangle define

$$N_E(T_C) = \left|\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}\right|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types)**.** *Every C triangle has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \le 2.$$

For the CE1/CE2 types, Proposition 2.11 below records the additional fact that T_C contains exactly one radial midpoint.

2.1.3. *Vertex types and the T3-like normalization.* For the closure T_i of an original open V triangle U_i, let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|.$$

We use the names

type	(o,n)
Vd0	$(1,0),(2,0),(3,0)$
Vd1	$(1,1)$
Vd2	$(1,2)$
T3-like	$(2,1)$.

Lemma 2.4 (Local wedge). *Every original V triangle has $1 \le o \le 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one of its vertices belongs to H; if both $\mathcal{H}^1(T_i \cap r_{i-1}) > 0$ and $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$, at least two of its vertices belong to H.*

Proposition 2.5 (Exhaustive vertex types). *Every original V triangle has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.6 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V_i-triangle. There is a translate $\widehat{T}$ of T, with the same orientation and side length, such that*

$$V_i \in \operatorname{relint}(a\ side\ of\ \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $\widehat{T}$ remains of type $(o,n) = (2,1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open triangle.

2.1.4. *Reach coordinates, supercritical V triangles, and gaps.* For the original V triangle U_i, define its maximal incoming, outgoing, and radial reaches by

$$A(U_i) = \sup_{\substack{t\in[0,1] \\ V_i + t(V_{i-1}-V_i) \in U_i}} t,$$
$$B(U_i) = \sup_{\substack{t\in[0,1] \\ V_i + t(V_{i+1}-V_i) \in U_i}} t,$$
$$C(U_i) = \sup_{\substack{t\in[0,1] \\ V_i + t(O-V_i) \in U_i}} t.$$

Closure does not change the suprema, so

$$A(U_i) = A(T_i), \qquad B(U_i) = B(T_i), \qquad C(U_i) = C(T_i).$$

For compact indexed formulas, put

$$A_i := A(U_i) = A(T_i), \qquad B_i := B(U_i) = B(T_i), \qquad C_i := C(U_i) = C(T_i).$$

The closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed reach lower bounds whenever actual and selected values occur together. A realizing triangle then satisfies

$$A_i \ge a_i, \qquad B_i \ge b_i, \qquad C_i \ge c_i.$$

An actual V triangle is supercritical if $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|. \tag{2}$$

This definition never counts a smaller reach pair in place of the actual V triangle.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \le x \le 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \le 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention retains the actual open traces; a point missing from both open triangles is not erased merely because both closures contain it.

Lemma 2.7 (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident V triangle traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center triangle. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two.*

Lemma 2.8 (T3-like V triangles are nonsupercritical)**.** *Every original T3-like V triangle satisfies*

$$A_i + B_i \le 1. \tag{3}$$

In particular, it is not counted by N_+.

We state the precise passage from maximal reaches to strict handoff lower bounds. Its hypothesis that the six V triangles themselves cover the boundary is part of the statement.

Proposition 2.9 (Strict boundary handoffs)**.** *Assume that $U_0, \ldots, U_5$ cover ∂H. On each edge choose*

$$\xi_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - \xi_{i-1}, \qquad b_i = \xi_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i, and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

1. *if exactly one actual V triangle, indexed by p, is supercritical, every strict selection has exactly one selected supercritical V triangle, also indexed by p;*
2. *if at least two actual V triangles are supercritical, some simultaneous strict selection has at least two selected supercritical V triangles.*

2.1.5. *Midpoint forcing.* We first need a local fact for V triangles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Lemma 2.10 (Self-midpoint obstruction)**.** *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

Then $a+b \le 1$. Consequently, for every V triangle,

$$M_i \in T_i \quad \Longrightarrow \quad A_i + B_i \le 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

The corresponding center fact is exact and uses the interior condition at O.

Proposition 2.11 (CE1/CE2 exactly-one-midpoint property)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$. If T_C is CE1 or CE2, equivalently if it has positive-length intersection with a boundary edge of H, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Remark 2.12. The hypothesis $O \in \operatorname{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1.

Proposition 2.13 (Exhaustive structural reduction)**.** *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in Table 1, and its maximal reaches determine exactly one row of that table.*

2.2. **Signed CE1/CE2 Reductions.** This section gives the signed center model used by both later methods. The propagation interface depending on that model is deferred until after the transfer calculus in Section 4.1.

2.2.1. *One signed center normal form.*

Proposition 2.14 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O \in \operatorname{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$0 < R < 1, \qquad W = 1 - R,$$
$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the C triangle is

$$\boxed{\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \\ F_2 &= W + \delta - b + Ra, \end{aligned}} \qquad T_C = \{F_0, F_1, F_2 \ge 0\}. \tag{4}$$

Define

$$\Delta_R = P - \alpha - W\delta, \qquad \Delta_L = P - R\alpha - \delta. \tag{5}$$

Then

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{6}$$

and the possible companion trace is

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R+\alpha\right] \quad \textit{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{7}$$

The normalization requires $\Delta_R > 0$, and

$$T_C \textit{ is CE1} \iff \Delta_L \le 0, \qquad T_C \textit{ is CE2} \iff \Delta_L > 0. \tag{8}$$

The six center exits, measured from O, are

$$\boxed{\begin{array}{lll} d_0^C = E-\alpha-\delta, & d_1^C = \dfrac{\delta}{R}, & d_2^C = \delta, \\ d_3^C = \min\left\{\dfrac{\alpha}{R}, \dfrac{\delta}{W}\right\}, & d_4^C = \alpha, & d_5^C = \dfrac{\alpha}{W}. \end{array}} \tag{9}$$

Moreover,

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}. \tag{10}$$

For closures of original open C triangles one has $\alpha, \delta > 0$.

Remark 2.15 (Affine-chart variables). If $\kappa_j = F_j(O)$ denotes the auxiliary side slack in the CE1 affine chart, take

$$\lambda = R, \qquad s = \frac{k}{R}, \qquad t = W+\delta, \qquad \kappa_0 = \alpha, \qquad \kappa_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \qquad \nu = R+\alpha, \qquad y = \frac{k}{R}, \qquad v = W+\delta.$$

If $S = x+y$ and $D = \sqrt{x^2+xy+y^2}$, then

$$S = \frac{k}{RW}, \qquad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu+v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 2.14.

2.2.2. *Boundary contribution and diameter transfer.*

Corollary 2.16 (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{11}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Lemma 2.17 (Adjacent-edge diameter transfer)**.** *For $0 \le q \le 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here the argument-free symbol M_0 continues to denote the radial midpoint; $M_0(q)$ denotes the envelope map. If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then

$$b \le M_0(q).$$

The map M_0 is strictly decreasing. For $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \qquad 1 - M_0(q) > \frac{q}{2}.$$

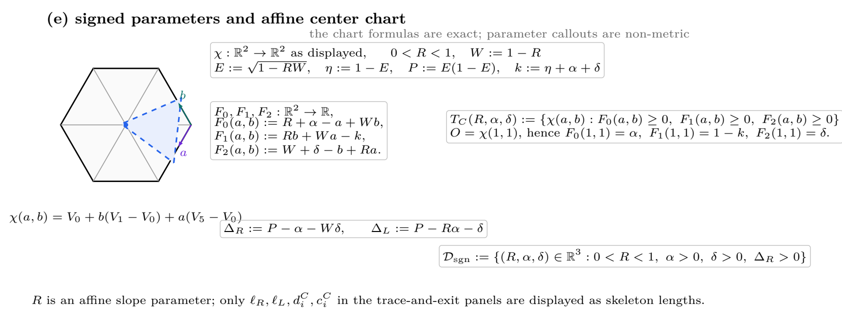


FIGURE 2. The affine center chart and the exact dependency structure of the signed parameters.

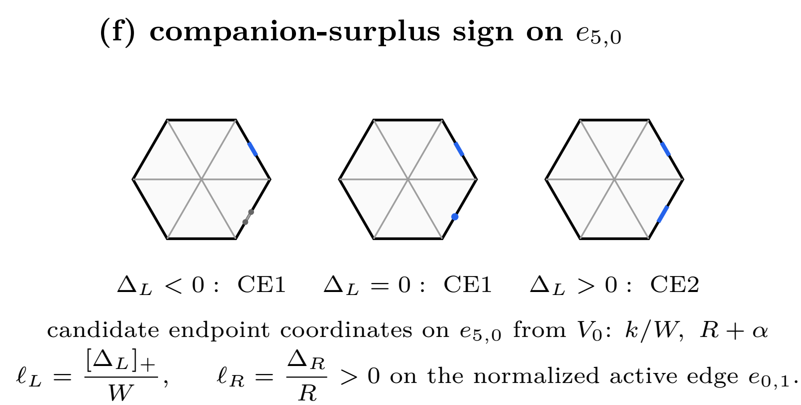
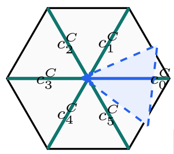


FIGURE 3. The exact sign test for the candidate companion trace, including the CE1 point-contact case.

(h) center exits and complementary radial reach lower bounds



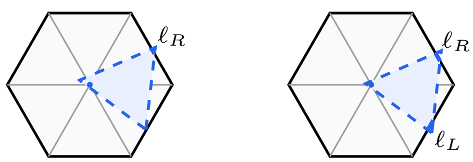
$d_0^C = E - \alpha - \delta$
$d_1^C = \delta/R$
$d_2^C = \delta$
$d_3^C = \min\{\alpha/R, \delta/W\}$
$d_4^C = \alpha$
$d_5^C = \alpha/W$

$c_i^C := 1 - d_i^C$
d_i^C: measured from O
c_i^C: measured from V_i

CE2 sample: $(d_0^C, \ldots, d_5^C) \approx (.8018, .0667, .04, .05, .03, .075)$.

FIGURE 4. The six exact center exits d_i^C and complementary radial lower bounds $c_i^C = 1 - d_i^C$.

(i) boundary-contribution bounds on the skeleton



$$\text{CE1}: \quad L_{\partial H}(T_C) = \ell_R \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}$$
$$\text{CE2}: \quad L_{\partial H}(T_C) = \ell_R + \ell_L < \frac{1}{2}$$

FIGURE 5. The exact CE1 and CE2 center-boundary contributions and their general bounds.

3. Trace-Length Bounds

The first mechanism compares the total one-dimensional trace available from the seven triangles with the length of either the perimeter ∂H or the full skeleton S. This section records the local trace estimates, the master perimeter deficit, the conditional skeleton theorem, and every routed application; their proofs are omitted.

3.1. **Detailed Trace-Length Estimates.** This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E, define

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

For the closures of a hypothetical open cover, the trace bound is

$$\mathcal{H}^1(E) \le L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{12}$$

All estimates in this section concern full traces and therefore apply to assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S, whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and V triangle hypotheses; no unconditional noncoverage assertion for S is made.

FIGURE 6. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

3.1.1. *Conditional skeleton trace caps.* We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 3.1 (Center skeleton cap). *If T_C is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Lemma 3.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle, so $V_i \in \mathrm{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \textit{for some } j \in \{i-1, i+1\},$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.3 (No-support skeleton cap). *If a V triangle has $A_i + B_i \le 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \le 2.$$

For the remaining cap, use the following selected supercritical cell. In local directions u, v meeting at $120°$, if a closed unit triangle contains $0, au, bv, c(u+v)$ with $0 \le a \le b \le 1$ and $a + b > 1$, then

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2-1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{13}$$

where c lies on the connected component containing $c = 0$.

Lemma 3.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.5 (A positive-support adjacent exceptional triangle is forced)**.** *Suppose T_C is CE1 or CE2 and, after symmetry, $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. If at least two V triangles are supercritical, then for a supercritical index $s \neq 0$ there is some $j \in \{s-1, s+1\}$ such that*

$$M_s \in U_j, \qquad \mathcal{H}^1(T_j \cap r_s) > 0.$$

3.1.2. *Vd1/Vd2 corner estimates.* We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition 3.6 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original Vd1 or Vd2 triangle at V_0, and let a, b be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x - (t+1)y \le a, \\ ty - (t+1)x \le tb, \\ tx + y \le d - a - tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d - a - tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d - a - tb - 1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d - a - tb - t. \end{array}$$

Intersecting with $[0,1]$ gives the actual traces. Moreover,

$$a + b < \frac{1}{2}. \tag{14}$$

Lemma 3.7 (Vd2 adjacent-midpoint cap)**.** *If an open role U_i has Vd2 closure $T_i = \overline{U_i}$ and T_i contains M_{i-1} or M_{i+1}, then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

3.1.3. *Boundary trace caps.* The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem 3.8 (Boundary trace table)**.** *For closures of original open triangles, the following bounds hold:*

triangle	*hypothesis*	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \dfrac{\sqrt{3}}{2} - \dfrac{3}{4}$
T_C	CE2	$< \dfrac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \dfrac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \dfrac{1}{2}$
T_i	*T3-like*	$< 1.$

3.1.4. *Final length contradictions.*

Proposition 3.9 (Vd2 adjacent-midpoint branch)**.** *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 triangle, that triangle is T_i of type Vd2 and contains M_{i-1} or M_{i+1}, and all other V triangles are Vd0. Then the boundary cannot be covered.*

Proposition 3.10 (Branches closed by Method 1)**.** *Direct length-sum obstructions close the following routing entries and hybrid subcase:*

(1) *CE0, $N_+ = 0$;*
(2) *CE0, $N_+ = 1$, with a Vd1/Vd2 triangle;*
(3) *CE1/CE2 with $N_+ + N_{\mathrm{sp}} \geq 3$;*
(4) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 triangle;*
(5) *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 triangle and no T3-like triangle;*
(6) *the CE2 hybrid subcase in which the Vd2 triangle T_i contains M_{i-1} or M_{i+1}.*

3.2. **Common perimeter and skeleton budgets.** The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then give the direct CE1/CE2 skeleton count used by the routing table.

Lemma 3.11 (Master perimeter deficit)**.** *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C, n supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \ldots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{15}$$

then the seven triangles do not cover ∂H.

Lemma 3.12 (Small center slacks in the one-Vd1/Vd2 branch)**.** *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four* *nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$\boxed{\alpha+\delta<\frac{1}{24},\qquad \alpha+\delta<\frac{\min\{R,W\}}{6}.} \tag{16}$$

Lemma 3.13 (CE2 initial endpoints dominate the total slack)**.** *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T=\alpha+\delta,\qquad x=\frac{\eta+T}{W},\qquad y=\frac{\eta+T}{R}.$$

Then

$$\boxed{T<\frac{1}{2}\min\{x,y\}.} \tag{17}$$

Theorem 3.14 (Direct CE1/CE2 skeleton count)**.** *If a CE1/CE2 branch satisfies $T_C \cap \{M_0,\ldots,M_5\} = \{M_0\}$ and*

$$N_+ + N_{\mathrm{sp}} \geq 3, \tag{18}$$

then the seven triangles do not cover the full skeleton S.

Corollary 3.15 (Length branches closed by the common budgets)**.** *The following branches are impossible.*

1. *CE0, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
2. *$N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;*
3. *CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
4. *CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;*
5. *CE2, $N_+ = 1$, with a Vd2 triangle T_i containing M_{i-1} or M_{i+1};*
6. *CE1 or CE2 with $N_+ + N_{\mathrm{sp}} \geq 3$;*
7. *CE1 or CE2 with $N_+ \geq 2$.*

4. Boundary-Reach Propagation

The second mechanism converts a backward boundary reach and a radial reach into an upper bound for the forward reach of the same V triangle. Coverage of the next edge then produces a lower bound for the next backward reach. This section records the common equilateral-enclosure calculus, center-assisted path budget, exceptional T3-like and Vd placements, and the branchwise propagation statements. Formalization-only registries remain repository verification objects and are not printed here.

4.1. **Universal Enclosure and Propagation Calculus.** This section collects the geometry shared by the propagation and direct-witness arguments. The same equilateral enclosure functional governs the local admissible set of Method 2 and the forced witness set of Method 4. The transfer statements below distinguish three situations that must not be conflated: a center-free edge, an edge with a specified center interval, and a free strict-supercritical outgoing envelope.

4.1.1. *The equilateral enclosure gauge.*

Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact set $K \subset \mathbb{R}^2$, put

$$h_K(n) = \max_{x \in K} \langle x, n \rangle$$

and

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{19}$$

Proposition 4.1 (Equilateral enclosure gauge)**.** *The number $\Lambda(K)$ is the least side length of a closed equilateral triangle containing K. It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a finite polygonal hull, a minimizing normal may be chosen perpendicular to one of its hull edges.*

For the anchor hull

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\},$$

where u and v are the two unit boundary directions meeting at $120°$, the admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \le 1\}. \tag{20}$$

The corresponding exact admissible-set statement is recorded in Section 4.6.

4.1.2. *One equilateral radical.*

For $0 \le x \le 1$, define

$$\omega(x) = \sqrt{1 - x + x^2}, \qquad \sigma(x) = 1 - \omega(x). \tag{21}$$

In the signed C-triangle calculus,

$$E = \omega(R), \qquad \eta = \sigma(R).$$

4.1.3. *Forward envelope and nonsupercritical propagation.*

For $0 \le a, c \le 1$, define the raw forward envelope

$$M_c(a) = \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}. \tag{22}$$

Its zero-radial specialization, introduced geometrically in Lemma 2.17, is

$$M_0(a) = \frac{-a + \sqrt{4 - 3a^2}}{2}. \tag{23}$$

The exact nonsupercritical forward cap and the following-edge propagation map are

$$\overline{M}_c(a) = \begin{cases} 1 - a, & 0 \le c \le 1/2, \\ M_c(a), & 1/2 < c \le 1, \end{cases} \qquad \Phi_c(a) = 1 - \overline{M}_c(a). \tag{24}$$

Equivalently,

$$\overline{M}_c(a) = \min\{M_c(a), 1 - a\}.$$

Lemma 4.2 (Midpoint threshold)**.** *For $0 \le a \le 1$,*

$$M_c(a) \le 1 - a \qquad \left(\frac{1}{2} \le c \le 1\right), \qquad M_{1/2}(a) = 1 - a.$$

Consequently $\Phi_c(a) \ge a$ whenever $c \ge 1/2$.

Lemma 4.3 (Raw envelope and nonsupercritical cap)**.** *Let T be an actual V triangle satisfying $A(T) \geq a$ and $C(T) \geq c$, and let T_{next} be the next incident actual V triangle. Then*

$$B(T) \leq M_c(a).$$

If no center interval intervenes on the next edge, then

$$A(T_{\mathrm{next}}) \geq 1 - M_c(a).$$

If T is nonsupercritical, then

$$B(T) \leq \overline{M}_c(a) \tag{25}$$

and, on a center-free next edge,

$$A(T_{\mathrm{next}}) \geq \Phi_c(a) \geq a. \tag{26}$$

Proposition 4.4 (Zero-radial warning)**.** *For $0 < a < 1$,*

$$M_0(a) > 1 - a, \qquad 1 - M_0(a) < a.$$

Thus a nonsupercritical zero-radial handoff uses the cap $\overline{M}_0(a) = 1 - a$, not the raw envelope $M_0(a)$.

4.1.4. *Center-assisted propagation and the free envelope.*

Let $J \subseteq [0, 1]$ be empty or a closed interval. For $0 \leq p \leq 1$, put

$$e_J(p) = \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup J\}, \qquad \mathcal{R}_J(p) = 1 - e_J(p). \tag{27}$$

If $J = [L, U]$, then

$$\mathcal{R}_{[L,U]}(p) = \begin{cases} 1 - p, & p < L, \\ 1 - \max\{p, U\}, & p \geq L, \end{cases} \qquad \mathcal{R}_\varnothing(p) = 1 - p. \tag{28}$$

The map is nonincreasing in p and under enlargement of J.

Lemma 4.5 (Generalized boundary handoff)**.** *Suppose a unit boundary edge is covered by a left trace contained in $[0, p]$, a C triangle trace J, and a right trace contained in $[1 - q, 1]$. Then*

$$q \geq \mathcal{R}_J(p).$$

For an actual V triangle satisfying $A(T) \geq a$ and $C(T) \geq c$,

$$A(T_{\mathrm{next}}) \geq \mathcal{R}_J(M_c(a)). \tag{29}$$

If T is nonsupercritical, then

$$A(T_{\mathrm{next}}) \geq \mathcal{R}_J(\overline{M}_c(a)). \tag{30}$$

For $J = \varnothing$, the latter is exactly $A(T_{\mathrm{next}}) \geq \Phi_c(a)$.

For $0 \leq c < 1/2$, define

$$M_c^{\sup} = \sup\{M_c(a) : M_c(a) > 1 - a\} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}. \tag{31}$$

The supremum is not attained. Hence an actual strict-supercritical V triangle T with $C(T) \geq c$ satisfies

$$B(T) < M_c^{\sup}. \tag{32}$$

If, in addition, its outgoing edge is center-free and T_{next} is the next incident actual V triangle, then coverage gives

$$A(T_{\mathrm{next}}) > 1 - M_c^{\sup}. \tag{33}$$

At $c = 1/2$ the strict feasible set is empty. When convenient we write $M^{\sup}_{1/2} = 1/2$ only for the continuous extension of the displayed formula, never for an attained strict supremum.

4.1.5. *Relaxed composition and path budgets.*

For maps listed in geometric V triangle order, write

$$[\Psi_1 \mid \cdots \mid \Psi_r](x) = (\Psi_r \circ \cdots \circ \Psi_1)(x). \tag{34}$$

The leftmost slot acts first. Write $\mathrm{I}(x) = x$.

Lemma 4.6 (Relaxed composition)**.** *Suppose propagated reach lower bounds satisfy*

$$x_j \geq \Psi_j(x_{j-1}) \qquad (1 \leq j \leq r),$$

where every Ψ_j is nondecreasing. If $\underline{\Psi}_j \leq \Psi_j$, then

$$x_r \geq [\underline{\Psi}_1 \mid \cdots \mid \underline{\Psi}_r](x_0).$$

Lemma 4.7 (Boundary-path budget)**.** *Let $E_0, \ldots, E_k$ be consecutive boundary edges and let $T_1, \ldots, T_k$ be the triangles at their intermediate vertices. Suppose:*

(1) *on E_0, the component covered from the endpoint opposite T_1 by all triangles other than T_1 has extent at most u;*
(2) *on E_k, the analogous component opposite T_k has extent at most v;*
(3) *on every middle edge, no triangle other than the two incident path triangles has positive-length trace;*
(4) *the chain is covered.*

Then

$$\sum_{j=1}^{k} (A_j + B_j) \geq k + 1 - u - v. \tag{35}$$

Consequently, if all k internal V triangles are nonsupercritical, the chain is impossible whenever $u + v < 1$.

4.1.6. *Low roots, selected Q_+ curves, and thresholds.*

For the high-radial maps define

$$e(d) = \frac{1-d}{2}\left(1 - \sqrt{4(1-d)^2 - 3}\right) \qquad \left(0 < d < 1 - \frac{\sqrt{3}}{2}\right), \tag{36}$$

and, for the low-radial order transition,

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}} \qquad \left(\frac{1}{2} < c \leq \frac{\sqrt{3}}{2}\right). \tag{37}$$

Proposition 4.8 (Universal selected-Q_+ curve)**.** *Fix $0 < d < 1/2$, put $c = 1 - d$, and suppose an input p belongs to a selected Q_+ arc of Φ_c. Write*

$$q = \Phi_c(p), \qquad \nu = q - p, \qquad x = \frac{q-d}{1-d}.$$

Then

$$\nu = \sigma(x) = 1 - \sqrt{1 - x + x^2}, \tag{38}$$

and

$$q = d + (1-d)x, \qquad p = d + (1-d)x - \sigma(x). \tag{39}$$

Consequently the selected arc is increasing and strictly concave.

The corresponding chord bounds are

$$\Phi_{1-d}(p) \geq p + \frac{d}{e(d)-d}(p-d), \tag{40}$$

and, with $t = h(1-d)$,

$$\Phi_{1-d}(p) \geq p + \frac{1-2t}{t-d}(p-d). \tag{41}$$

For every coefficient λ valid on the relevant selected arc, write

$$\Phi^{\lambda}_{1-d}(p) := p + \lambda(p-d) \leq \Phi_{1-d}(p). \tag{42}$$

For $0 < d < 1 - \sqrt{3}/2$, the exact high-reach lower-bound threshold is

$$z > e(d) \quad \Longrightarrow \quad \Phi_{1-d}(z) \geq 1 - e(d). \tag{43}$$

Define

$$\Phi^{\mathrm{th}}_{1-d}(z) = \begin{cases} z, & z \leq e(d), \\ 1-e(d), & z > e(d), \end{cases} \qquad \Phi^{\mathrm{th}}_{1-d} \leq \Phi_{1-d}. \tag{44}$$

Lemma 4.9 (Threshold routing)**.** *Let $z_j = \Psi_j(z_{j-1})$, where every Ψ_j is nondecreasing and extensive. If $\Psi_k = \Phi_{1-d}$ and $z_{k-1} > e(d)$, then $z_j \geq 1 - e(d)$ for all $j \geq k$. If a chain contains later occurrences of Φ_{1-d_1} and Φ_{1-d_2} and*

$$z_0 > e(d_1), \qquad \min\{e(d_1), e(d_2)\} < Q,$$

then at least one of those two V triangles forces every subsequent output above $1-Q$.

4.2. **The local propagation interface.** For the high-radial parameters used below, M_c is the raw outgoing envelope and $\Phi_c = 1 - \overline{M}_c$ is the following-edge propagation map. The latter is nondecreasing and extensive:

$$\Phi_c(z) \geq z. \tag{45}$$

The exact reflection duality is

$$\Phi_c(a) \leq z \quad \Longleftrightarrow \quad \Phi_c(1-z) \leq 1-a. \tag{46}$$

Let a nonsupercritical V triangle have maximal reaches (A_i, B_i, C_i) with $A_i \geq z$ and $C_i \geq c_i > 1/2$. If the next boundary handoff gives $A_{i+1} \geq 1 - B_i$, then

$$A_{i+1} \geq \Phi_{c_i}(z). \tag{47}$$

Every composition below abbreviates repeated use of this statement.

4.3. **The common one-gap interface.**

Proposition 4.10 (Common five-V triangle one-gap reduction)**.** *Assume every V triangle is Vd0, $N_+ = 1$, and the normalized right center trace contains a boundary gap, possibly a singleton. In the CE2 case assume the companion trace contains no boundary gap. Put*

$$X = R - \delta, \qquad h_0 = \frac{k}{2R}, \qquad m_3 = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

and

$$c_1 = 1 - \frac{\delta}{R}, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m_3,$$
$$c_4 = 1 - \alpha, \quad c_5 = 1 - \frac{\alpha}{W}, \quad c_0 = k.$$

Let

$$z_0 = X, \qquad z_i = \Phi_{c_i}(z_{i-1}) \quad (1 \le i \le 5),$$

and $Z = z_5$. *Then the maximal reaches satisfy*

$$A_0 \ge Z, \qquad A_0 \le M_k\left(\frac{k}{R}\right) < 1 - h_0. \tag{48}$$

The corresponding scalar target is

$$[\Phi_{1-\alpha} \mid \Phi_{1-m_3} \mid \Phi_{1-\delta}](h_0) > W + \delta. \tag{49}$$

The CE1 sign $\Delta_L \le 0$ is completed by the selected-Q_+ relaxation in Proposition 4.26. We now give the shorter CE2 scalar clause.

Lemma 4.11 (CE2 total slack and two thresholds)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$, *and put*

$$T = \alpha + \delta, \qquad Q = \frac{\eta + T}{2R}.$$

Then

$$T < \frac{\eta}{E}, \tag{50}$$

and

$$\min\{e(\alpha), e(\delta)\} < Q. \tag{51}$$

Proposition 4.12 (Short CE2 one-gap completion)**.** *Under the CE2 hypotheses of Proposition 4.10, the one-gap state is impossible in either orientation.*

Remark 4.13 (Why the CE1 scalar clause remains separate). The geometric one-gap interface is common, but (51) is not valid on the full CE1 sign domain. Thus the merger stops at Proposition 4.10; the CE1 selected-Q_+ clause and the CE2 threshold clause require different scalar arguments for the same target (49).

4.4. **Exceptional Local Profiles and Vd Placements.** This subsection records the local profile estimates used by the one-special-triangle branches; their proofs are omitted.

Use the strict-supercritical envelope $M_c^{\sup}$ from (31); its explicit formula is strictly decreasing for $0 \le c < 1/2$. At $c = 1/2$ we use only the continuous extension $M_{1/2}^{\sup} = 1/2$. The strict feasible set is empty there, so this endpoint value is never described as an attained supremum.

4.4.1. *A global quarter radial envelope.*

For $(p, h) \in [0, 1]^2$, let $c_{\max}(p, h)$ be the maximal radial coordinate such that (p, h, c) belongs to the local admissible set.

Lemma 4.14 (Quarter radial envelope)**.** *If*

$$0 \le h \le p, \qquad p + h < 1,$$

then

$$c_{\max}(p, h) \le 1 - \frac{h}{4}. \tag{52}$$

The inequality is strict on the selected $\mathcal{A}_T$ component.

4.4.2. *The rational T3-like adjacent-exception profile.*

Lemma 4.15 (T3-like adjacent-exception profile)**.** *Let a translated T3-like triangle at V_0 contain M_1, have boundary trace $[0, a]$ on $e_{5,0}$, and have trace $[c, u]$ on r_1 measured from V_1 toward O. Then*

$$a \le 1 - M_c^{\sup}, \qquad \frac{a}{a+1-u} \le 1 - M_c^{\sup}. \tag{53}$$

4.4.3. *The Vd1 adjacent-exception profile.*

Lemma 4.16 (Vd1 adjacent-exception profile)**.** *Let a Vd1 triangle at V_0 contain M_1, have boundary trace $[0, a]$ on $e_{5,0}$, and have trace $[c, u]$ on r_1. Assume it has no positive support on r_5. Then*

$$a \le 1 - M_c^{\sup}, \qquad \frac{a}{a+1-u} \le 1 - M_c^{\sup}. \tag{54}$$

4.4.4. *A common adjacent-exception theorem.*

Lemma 4.17 (Common adjacent-exception obstruction)**.** *Let a local adjacent exceptional triangle T_0 have trace $[0, a]$ on $e_{5,0}$ and trace $[c, u]$ on r_1, measured from V_1 toward O. Put*

$$\varepsilon = 1 - u > 0, \qquad \Theta = \frac{a}{a+\varepsilon}.$$

Assume

$$a \le 1 - M_c^{\sup}, \qquad \Theta \le 1 - M_c^{\sup}, \tag{55}$$

and assume radial isolation forces the center to cover the O-side gap of length ε. If T_1 is the unique strict supercritical V triangle and T_2, T_3, T_4, T_5 are nonsupercritical, then the perimeter and r_1 are not covered.

Corollary 4.18 (Adjacent T3-like/Vd1 rescue)**.** *The common adjacent-exception lemma applies both to the exactly-one-T3-like branch and to a normalized Vd1 triangle T_i containing M_{i-1} or M_{i+1}.*

4.4.5. *Adjacent Vd1/Vd2 placement.*

Lemma 4.19 (Adjacent Vd1/Vd2 placement)**.** *Assume a reduced CE2 candidate with $T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}$ has $N_+ = 1$, V triangle T_0 uniquely supercritical, V triangle T_1 the unique Vd1/Vd2 V triangle, and V triangles T_2, T_3, T_4, T_5 nonsupercritical Vd0. Then the perimeter together with r_2 is not covered. The reflected T_5 placement is also impossible.*

4.4.6. *Nonadjacent Vd1/Vd2 placements.*

Lemma 4.20 (Nonadjacent Vd1/Vd2 placements)**.** *Under the same reduced CE2 branch, let the unique Vd1/Vd2 V triangle be T_τ with $\tau \in \{2, 3, 4\}$. Then the perimeter together with r_τ is not covered.*

4.4.7. *Placement interfaces.*

The local profile and radial-separation lemmas above are assembled exactly once in Section 4.10. This subsection contains no independent CE2 one-Vd placement assembly.

4.5. **Propagation roadmap and the T3 endpoint state.** The following statements collect eight geometric outputs: the one- and two-gap endpoint inequalities, the CE1 and CE2 return inequalities, the support-isolated T3 endpoint inequality, the common adjacent-exception obstruction, and the two Vd separation statements. Their labels are 4.33, 4.34, 4.26, 4.12, 4.31, 4.18, 4.19, and 4.20. The detailed proofs are omitted.

Definition 4.21 (Support-isolated T3-like endpoint state)**.** Assume T_C is CE1 or CE2, $N_+ = 0$, no V triangle is Vd1 or Vd2, and, after normalization and reflection, T_0 is T3-like with $\mathcal{H}^1(T_0 \cap r_1) > 0$. Apply the closed-trace domination of Proposition 2.6 and retain the symbol T_0 for the resulting trace-dominating majorant. The support-isolation hypotheses are

$$\mathcal{H}^1(T_j \cap r_1) = 0 \quad (j \notin \{0,1\}), \qquad \mathcal{H}^1(T_j \cap r_5) = 0 \quad (j \neq 5).$$

Parameterize $e_{0,1}$ and $e_{5,0}$ from V_0, and r_1, r_5 from O. Write

$$T_0 \cap e_{0,1} = [0, p_1], \qquad T_0 \cap e_{5,0} = [0, p_5],$$

$$J_1 = T_C \cap e_{0,1}, \qquad J_5 = \begin{cases} T_C \cap e_{5,0}, & T_C \text{ is CE2}, \\ \varnothing, & T_C \text{ is CE1}. \end{cases}$$

In CE1 a possible closure-only point contact is discarded: it has zero trace, and the original open C triangle supplies neither open coverage nor a positive boundary interval there. Write

$$T_C \cap r_i = [0, \gamma_i] \quad (i \in \{1,5\}), \qquad T_0 \cap r_1 = [1-u, 1-c].$$

With the residual operator $\mathcal{R}_J$ of (27), define

$$(a_5^*, c_5^*) = \left(\mathcal{R}_{J_5}(p_5), 1-\gamma_5\right),$$

$$(a_1^*, c_1^*) = \begin{cases} \left(\mathcal{R}_{J_1}(p_1), c\right), & 1-u \le \gamma_1 \le 1-c, \\ \left(\mathcal{R}_{J_1}(p_1), 1-\gamma_1\right), & \text{otherwise}. \end{cases}$$

For $i \in \{1,5\}$ the endpoint capacity is $\overline{M}_{c_i^*}(a_i^*)$; it is the direct nonsupercritical value $1 - a_i^*$ when $c_i^* \le 1/2$. If A_i, C_i denote the actual maximal boundary and radial reaches of T_i, then the displayed residual quantities are the selected lower bounds

$$a_i^* \le A_i, \qquad c_i^* \le C_i \qquad (i \in \{1,5\}).$$

Proposition 4.22 (Common CE2 two-gap application)**.** *Assume both CE2 center traces contain boundary gaps and $T_1, \dots, T_5$ are nonsupercritical Vd0 triangles. Then the perimeter is not covered. No hypothesis on the criticality of T_0 is required.*

4.6. **Detailed Local Reach Calculus.** This section develops the local calculus for configurations in which a coarse length sum does not by itself retain enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial lower bounds; an exact local map propagates these data through nonsupercritical V triangles; and the returning reach lower bound exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by (A_i, B_i, C_i), while lowercase letters denote prescribed reach lower bounds. The complete paper establishes all estimates analytically; this edition retains their statements.

4.6.1. *The exact local feasible set.* Retain $K(a,b,c)$, the admissible set $\mathcal{A}$, and its raw envelope $M_c(a)$ from (20) and (22). The variables a,b,c are selected reach lower bounds; actual maximal reaches remain uppercase.

Proposition 4.23 (Exact admissible set and radial envelope)**.** *Put*

$$s = a+b, \qquad d = a^2+ab+b^2, \qquad m = \min\{a,b\}, \qquad M = \max\{a,b\},$$

and

$$q = s^4 - s^2 + ab.$$

Define

$$\begin{aligned} F_L(a,b,c) &= c^4 - c^2 + mc - m^2, \\ F_T(a,b,c) &= (s^2-1)c^2 + Mc - M^2, \\ F_S(a,b,c) &= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2. \end{aligned}$$

Then $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$*, where*

$$\begin{aligned} \mathcal{A}_L &= \{d \le 1, s \le 1, q \le 0, F_L \le 0\}, \\ \mathcal{A}_T &= \{d \le 1, s \le 1, q \ge 0, F_T \le 0, c \le 2M\}, \\ \mathcal{A}_S &= \{d \le 1, s > 1, F_S \le 0, c \le \tfrac{1}{2}\}. \end{aligned}$$

All variables in these three sets are also constrained to $[0,1]$*.*

For $d \le 1$*, the maximal radial lower bound is*

$$c_{\max}(a,b) = \begin{cases} 1, & a=b=0, \\ C_L(m), & s \le 1, q \le 0, (a,b) \ne (0,0), \\ C_T(m,M), & s \le 1, q > 0, \\ C_S(m,M), & s > 1, \end{cases} \tag{56}$$

where $C_L(m)$ *is the unique root in* $[\sqrt{3}/2, 1]$ *of*

$$z^4 - z^2 + mz - m^2 = 0,$$

and

$$C_T(m,M) = \frac{2M}{1+\sqrt{4s^2-3}}, \qquad C_S(m,M) = \frac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}.$$

If $d > 1$*, the radial envelope is undefined. At* $q = 0$ *the two subcritical formulas agree and equal* s*.*

4.6.2. *Branchwise nonsupercritical output.*

Proposition 4.24 (Exact midpoint split)**.** *For every* $a, c \in [0,1]$*,*

$$\overline{M}_c(a) = \min\{M_c(a), 1-a\}.$$

If an actual triangle has reaches (A,B,C) *with*

$$A \ge a, \qquad C \ge c, \qquad A+B \le 1,$$

then

$$B \le \overline{M}_c(a). \tag{57}$$

Moreover,

$$\overline{M}_c(a) = 1-a \quad (c \le 1/2), \qquad \overline{M}_c(a) = M_c(a) \quad (c > 1/2).$$

4.6.3. *The exact high-radial map.* For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and use the order-transition function $h(c)$ from (37). Also define

$$Q_-(a,c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$
$$Q_+(a,c) = \frac{2(1-a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1-c^2)(1-a^2)c^2}}.$$

Proposition 4.25 (Four-branch high-radial map)**.** *For $1/2 < c \leq \sqrt{3}/2$,*

$$\overline{M}_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ Q_+(a,c), & 1-c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a,c), & h(c) < a < c, \\ 1-a, & c \leq a \leq 1. \end{cases} \tag{58}$$

For $\sqrt{3}/2 < c < 1$,

$$\overline{M}_c(a) = \begin{cases} 1-a, & 0 \leq a \leq 1-c, \\ Q_+(a,c), & 1-c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a,c), & r(c) \leq a < c, \\ 1-a, & c \leq a \leq 1. \end{cases} \tag{59}$$

Thus there are exactly four branches

$$\text{Lin}, \qquad \text{Const}, \qquad Q_-, \qquad Q_+.$$

At $a = \ell(c)$ in (59), $\overline{M}_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the Q_+ quadratic is not feasible because it violates $c \leq 2\max\{a, \overline{M}_c(a)\}$.

The maps $\overline{M}_c$ are nonincreasing, the maps $\Phi_c = 1 - \overline{M}_c$ are nondecreasing and extensive, and

$$\Phi_c(a) \leq z \iff \Phi_c(1-z) \leq 1-a. \tag{60}$$

Put

$$\mu = \frac{2\sqrt{3} - 3}{4},$$

and, for $0 < X < 1 - \sqrt{3}/2$, write

$$e(X) = \ell(1 - X).$$

The root equation gives

$$\begin{aligned} X &< e(X) && \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right), \\ e(X) &< \frac{2X}{1-2X} && (0 < X \leq \mu), \\ e(X) &\leq 2X + 5X^2 && (0 < X \leq 1/8). \end{aligned} \tag{61}$$

Moreover,

$$a > e(X) \implies \Phi_{1-X}(a) \geq 1 - e(X) \qquad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right). \tag{62}$$

4.7. **Cyclic Propagation in the All-Vd0 Cases.**

4.7.1. *The exactly-one-supercritical all-*Vd0 *branch.* For a center interval, call the nonempty closed set missed by its two open endpoint triangles a *boundary gap*. It may be a singleton. If the maximal closed center trace is $[s, t]$, open coverage gives the weak bounds $B_i \geq s$ and $A_{i+1} \geq 1 - t$; thus none of the arguments below discards a singleton gap.

Proposition 4.26 (CE1 one-gap five-map obstruction)**.** *Assume that every V triangle is* Vd0*, $N_+ = 1$, and the C triangle is CE1. If its boundary trace contains a boundary gap, the triangles do not cover H.*

Proposition 4.27 (nonempty gap all-Vd0, $N_+ = 1$)**.** *If the C triangle is CE1 or CE2, every V triangle is* Vd0*, and $N_+ = 1$, then no cover of H can have a boundary gap in the actual open V triangle traces.*

4.8. **Nonsupercritical, T3-like, and Vd1/Vd2 Reach Branches.**

4.8.1. *The exactly-one-$T3$ branch.*

Proposition 4.28 (Exactly one T3-like V triangle)**.** *Assume the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is T3-like, and no triangle is Vd1 or Vd2. Then the triangles do not cover H.*

4.8.2. *The nonsupercritical CE1/CE2 boundary packages.* Figure 7 gives a common schematic for the package recorded below.

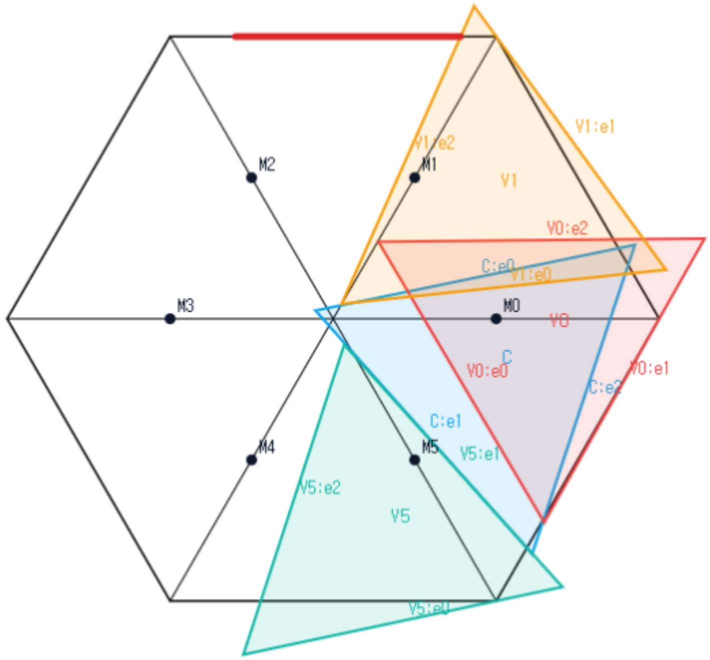


FIGURE 7. Schematic view of the common CE1/CE2, $N_+ = 0$, all-Vd0 boundary package, not to scale. The C triangle and three representative Vd0 vertex triangles are shown; the remaining three triangles are omitted for readability. The simultaneous placement is not a candidate cover, and no conclusion depends on visual inspection.

Proposition 4.29 (All-Vd0 skeleton-data obstruction)**.** *Let the C triangle be CE1 or CE2. Assume all six V triangles are Vd0,*

$$A_i + B_i \le 1 \qquad (i = 0, \dots, 5),$$

and the seven open triangles cover the full skeleton

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then no such configuration exists. Consequently the same distinguished-index data cannot cover all of H.

We next treat the additional T3-like V triangles in the $N_+ = 0$ branch. The following local facts also reappear later.

Lemma 4.30 (T3-like midpoint and side tradeoff)**.** *Let T_i be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition 2.6, suppose $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$ and M_{i+1} belongs to the dominated trace. If b is its boundary reach toward V_{i+1}, c its exit on r_i, and ℓ its entry on r_{i+1}, all measured from the relevant vertex, then*

$$b + c < 1, \qquad b + \ell > 1. \tag{63}$$

Consequently two T3-like traces T_i, T_{i+1} that cover each other's midpoint, with positive traces on r_{i+1} and r_i, respectively, cannot cover both segments $[V_i, M_i]$ and $[V_{i+1}, M_{i+1}]$ if no other triangle has positive trace there.

Lemma 4.31 (The support-isolated four-branch inequality)**.** *Assume the support-isolated T3-like endpoint state of Definition 4.21. Then the two endpoint V triangles satisfy*

$$\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1, \tag{64}$$

where a_1^*, a_5^* are the exact boundary inputs and c_1^*, c_5^* the exact radial lower bounds defined there.

Proposition 4.32 (The $N_+ = 0$ T3-like branch)**.** *Assume that T_C is CE1 or CE2, $N_+ = 0$, no V triangle is Vd1 or Vd2, and at least one V triangle is T3-like. Then the seven triangles do not cover H.*

4.8.3. *Two reusable high-radial endpoint inequalities.*

Lemma 4.33 (One-side endpoint loss)**.** *Let $0 < R < 1$, $S = \sqrt{1 - R + R^2}$, and let $s, u, w > 0$ satisfy $s + u + w = 1$ and $0 < u < R$. Put*

$$\delta = \frac{R}{1+S}, \qquad \gamma = u - \delta - \frac{R}{1-R}w,$$

and assume $\gamma > 0$. Set

$$c_1 = u/R, \qquad c_5 = 1 - \gamma.$$

Then

$$\overline{M}_{c_5}(s) + \overline{M}_{c_1}(u) < 1. \tag{65}$$

Lemma 4.34 (CE2 two-endpoint loss)**.** *Let $[x, U]$ and $[y, v]$ be the two exact CE2 traces in the legacy coordinates of Remark 2.15, where U denotes the endpoint called u there. Put*

$$p = 1 - U, \qquad q = 1 - v, \qquad r = \frac{x}{x+y}, \qquad w = \frac{y}{x+y},$$

and reflect if necessary so that $0 < w \le 1/2$ and $r = 1 - w$. Then

$$\overline{M}_{p/w}(p) + \overline{M}_{q/r}(q) < 1. \tag{66}$$

4.8.4. *The exactly-one-Vd1/Vd2 placements.* We now assume CE2, $N_+ = 1$, and exactly one V triangle is Vd1 or Vd2. The corner normal form of Proposition 3.6 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2 + t + 1}$,

$$x - (t+1)y \le a, \qquad ty - (t+1)x \le tb, \qquad tx + y \le d - a - tb. \tag{67}$$

In particular $a + b < 1/2$ for the Vd1/Vd2 triangle in this corner normal form.

Lemma 4.35 (Vd1 adjacent rescue)**.** *Assume an original open-triangle skeleton cover in the CE2 branch with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$, with T_0 the unique Vd1 V triangle, $M_1 \in T_0$, T_1 uniquely supercritical, $T_2, \dots, T_5$ nonsupercritical Vd0, and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0 \qquad (i = 1, \dots, 5).$$

Then the perimeter together with r_1 is not covered.

Lemma 4.36 (Vd1-pair replacement interface)**.** *In the remaining adjacent Vd1–supercritical placement, the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving full skeleton coverage.*

4.8.5. *Final boundary-propagation cases.*

Proposition 4.37 (Cases excluded by boundary propagation)**.** *Boundary-reach propagation gives the following exclusions:*

(1) *CE1 or CE2, $N_+ = 0$, all V triangles Vd0;*
(2) *CE1 or CE2, $N_+ = 0$, with at least one T3-like triangle and no Vd1/Vd2;*
(3) *CE1 or CE2, $N_+ = 1$, all V triangles Vd0, with at least one boundary gap;*
(4) *CE1 or CE2, $N_+ = 1$, exactly one T3-like triangle and no Vd1/Vd2;*
(5) *the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch with no T3-like triangle.*

All gap alternatives include singleton gaps.

4.9. **The Vd1–supercritical replacement.** The complete four-branch T3 endpoint statement is Lemma 4.31; its authenticated source crosswalk is kept in the paper source ledger rather than printed a second time. The next statement is the two-chart replacement needed for the Vd branch.

Proposition 4.38 (Vd1–supercritical two-chart replacement)**.** *Assume the CE2 branch in which an adjacent Vd1 triangle and the unique supercritical Vd0 triangle lie away from the V triangle at the center's unique midpoint. After reflection, write their base vertices as $V_\tau, V_{\tau+1}$. Assume the shared boundary edge is center-free, all other V triangles are nonsupercritical Vd0, and*

$$\mathcal{H}^1(T_j \cap r_\tau) = \mathcal{H}^1(T_j \cap r_{\tau+1}) = 0 \qquad (j \notin \{\tau, \tau+1\}).$$

Then the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving coverage of the full skeleton.

4.10. **Authoritative CE2 One-Vd Placement Assembly.** The following proposition is the CE2 exactly-one-Vd placement assembly used by the complete paper. It cites the placement lemmas and the two-chart replacement recorded above.

Proposition 4.39 (Complete CE2 exactly-one-Vd1/Vd2 assembly)**.** *Assume the center has type CE2, $N_+ = 1$, exactly one V triangle has type Vd1 or Vd2, and no T3-like triangle is present. Then the seven triangles do not cover H.*

5. Area-Loss Estimate

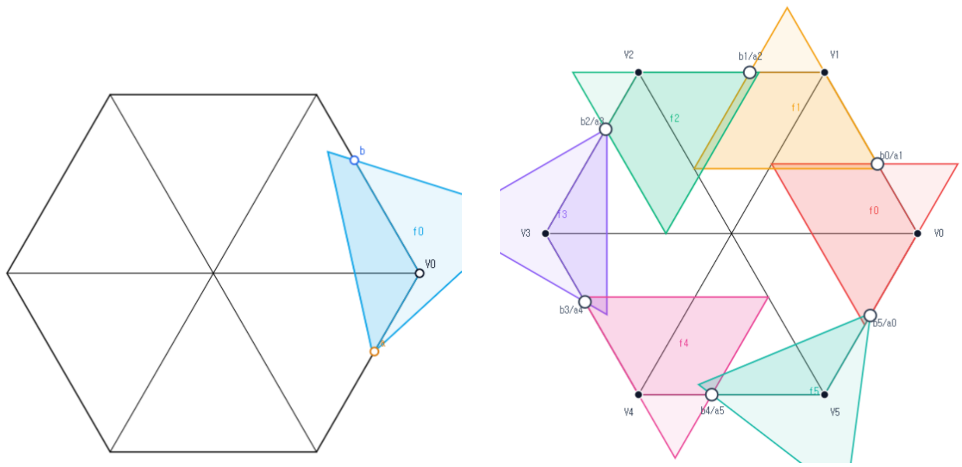
The third mechanism measures the normalized area forced outside H by a V triangle with prescribed boundary reaches. The local square-loss and T3-like estimates are stated first, followed by the two CE0 area-branch conclusions.

5.1. **Detailed Area-Loss Estimates.** This section records the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_\triangle = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_\triangle$. Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area (b) global retained areas

FIGURE 8. Schematic views of Method 3, not to scale. The local panel shows one V triangle with boundary reaches a, b, and the global panel shows the six cyclic triangles. Only the darker portions represent area retained inside H; the stated inequalities, rather than the picture, give the comparison.

5.1.1. *The local square-loss inequalities.* Fix a vertex V_i. Let P_a and P_b be the points at distances a and b from V_i on its incoming and forward boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b, put

$$\mathcal{L}(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b) = \max_T \frac{\operatorname{area}(T \cap H)}{A_\triangle}, \qquad \mathcal{L}(a,b) = 1 - f(a,b),$$

then $\mathcal{L}(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b, and increasing either reach lower bound can only decrease f.

Lemma 5.1 (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Theorem 5.2 (Unconditional local square loss)**.** *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T,*

$$\mathcal{L}(T) \geq \min(a,b)^2. \tag{68}$$

If $a + b > 1$, then

$$\mathcal{L}(T) \geq \max(a,b)^2. \tag{69}$$

Consequently the same inequalities hold for $\mathcal{L}(a,b) = 1 - f(a,b)$.

5.1.2. *The direct* T3-like *loss.* The next theorem uses the classification data (o, n) from Section 2.1: o counts triangle vertices outside H, and $n = n(T_i)$ is

$$\left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|.$$

Thus a T3-like triangle has $(o, n) = (2, 1)$.

Theorem 5.3 (Direct T3-like loss)**.** *Let T be a closed unit V_i-triangle of* T3-like *type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$a + b \leq 1. \tag{70}$$

Moreover, for $0 \leq m \leq 1/2$,

$$a, b \geq m \quad \Longrightarrow \quad \mathcal{L}(T) \geq 2m - 4m^2. \tag{71}$$

5.1.3. *The cyclic area certificates.* The selected handoffs of Proposition 2.9 have the form

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1,$$

with cyclic indices. V triangle i is selected supercritical exactly when $\xi_i > \xi_{i-1}$.

Lemma 5.4 (Cyclic area-loss certificate)**.** *Let the six actual V triangles realize selected handoffs*

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1.$$

Their total normalized loss is strictly greater than one if either

(1) *at least two selected V triangles are supercritical; or*
(2) *exactly one selected V triangle is supercritical, every triangle is* Vd0 *or* T3-like*, and at least one triangle is* T3-like*.*

Proposition 5.5 (Branches closed by area)**.** *The following branches are impossible:*

(1) *T_C is* CE0*, $N_+ \geq 2$, with arbitrary V triangle types;*
(2) *T_C is* CE0*, $N_+ = 1$, no triangle is* Vd1 *or* Vd2*, and at least one triangle is* T3-like*.*

6. Nine-Point Enclosure Obstruction

The fourth mechanism applies when all six V triangles are Vd0, the boundary is covered, and exactly one V triangle is supercritical. Six radial points and three asymmetric frontier points are forced into the C triangle. This section records the construction, support-arc, and final enclosure statements.

6.1. **Detailed Nine-Point Enclosure Argument.** This section records the exact frontier, witness, sign, and cap-overlap statements for the direct nine-point obstruction. The result is independent of the C-triangle class: all six V triangles are Vd0, they cover the full boundary, and exactly one actual boundary V triangle is supercritical. Six common radial points and three asymmetric points are forced into the C triangle.

6.1.1. *The finite caliper criterion.* We record the finite caliper criterion used below; its verification is omitted from this reading edition.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z \in S} \langle z, \mathsf{R}^k N \rangle. \tag{72}$$

Theorem 6.1 (Finite caliper certificate). *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon \mathsf{J}(q-p)\big) \le \frac{\sqrt{3}}{2} \|p - q\|. \tag{73}$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (73) *is a finite polynomial clause of degree at most four in the coordinates of A, B, x; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.*

6.1.2. *Corner-clipped AB-unions.* Let e_A, e_B be unit vectors with angle $120°$, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \ge 0\},$$

and define

$$\mathcal{R}(a,b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}. \tag{74}$$

The finite description in Theorem 6.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (74) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem 6.2 (Strict supercritical AB-union). *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}(a,b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{75}$$

where

$$\mathcal{D}_A = \{(u,v) : (u-a)^2 + v^2 - (u-a)v \le 1\},$$
$$\mathcal{D}_B = \{(u,v) : u^2 + (v-b)^2 - u(v-b) \le 1\},$$
$$\mathcal{H}_A = \{(u,v) : \alpha(u-a) + \beta v \le 0\},$$
$$\mathcal{H}_B = \{(u,v) : \gamma u + \delta(v-b) \le 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

6.1.3. *Exact-one handoffs and common reach bounds.*

Lemma 6.3 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.9 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \qquad b = \xi_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

one has

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{76}$$

and every selected V triangle obeys

$$a_i \geq p, \qquad b_i \geq q.$$

6.1.4. *Six directly forced radial points.* Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (76) gives

$$pq > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > 1 - m. \tag{77}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition 4.23. On the present subcritical domain, equation (56) becomes

$$c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{78}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma 6.4 (Direct radial forcing). *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{79}$$

Assume that every U_i is Vd0 *and that $U_C, U_0, \ldots, U_5$ cover H. Then, for every i,*

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$\mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \qquad \eta = h(1 - c_*). \tag{80}$$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a. Accordingly, set

$$\alpha = \frac{h(b+2a-bD)}{2\rho}, \qquad \beta = \frac{h(b-a+(a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-b+a+(a+b)D)}{2\rho}, \qquad \delta = \frac{h(2b+a-aD)}{2\rho},$$

where now $\rho = a^2+ab+b^2$ and $D = \sqrt{4\rho-3}$. The two line pieces of Theorem 6.2 are

$$L_-(\lambda) = (b-\beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a-\gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3}\right). \tag{81}$$

Put

$$E_- = (1-b, 2-b), \qquad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t,t), \qquad 1-a \le t \le b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \qquad 1-b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha+b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta+a\gamma)\mu + 3a^2 - 3a + 2. \tag{82}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha+b\beta) - \frac{2}{3}\sqrt{9(\alpha+b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta+a\gamma) - \frac{2}{3}\sqrt{9(\delta+a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma 6.5 (Moving adjacent-triangle signs)**.** *Throughout* (76),

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{83}$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3-2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P,t) = \mathsf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J,t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{84}$$

Under the reflection $(u,v,a,b) \leftrightarrow (v,u,b,a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}} = L_-(\lambda_-), \qquad Q_0^{\mathrm{loc}} = J, \qquad Q_+^{\mathrm{loc}} = L_+(\mu_+), \tag{85}$$

and by the displayed conversion from (u,v) to the plane.

Lemma 6.6 (Direct asymmetric forcing)**.** *Assume that* $U_C, U_0, \dots, U_5$ *cover* H*. Then*

$$Q_-, Q_0, Q_+ \in \operatorname{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C*.*

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas 6.4 and 6.6 therefore force the compact set

$$K_{\mathrm{wit}}(a,b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{86}$$

6.1.5. *Tangent reduction and exact mixed overlaps.* For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (82), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g_-'(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g_+'(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

Lemma 6.7 (Newton inner reduction)**.** *The points above have coordinates rational in* a, b, D *and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \operatorname{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \operatorname{conv}(K_{\mathrm{wit}})$*.*

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

6.1.6. *Ray order and four overlaps.*

Proposition 6.8 (Technical four-overlap verification)**.** *Assume* (76) *and* $c_* > 2/3$*. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA *occur in that cyclic order, and the four consecutive cap pairs*

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

6.2. **Support arcs and the enclosure conclusion.** Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C, the vector $W = C + \mathsf{R}A$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \qquad P_0 = B, \qquad P_+ = C, \qquad P_* = W, \qquad r_* = \eta,$$

and

$$\widehat{K} = \operatorname{conv}\big(\mathcal{D}_\eta \cup \{A, B, C\}\big).$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\},$$

and define the level-h, $h = \sqrt{3}/2$, support arc

$$\operatorname{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n\rangle + r \geq h\}.$$

Lemma 6.9 (Cyclic support-arc covering)**.** *Let K be compact and put*

$$\Sigma_K = K + \mathsf{R}K + \mathsf{R}^2K.$$

Suppose Σ_K contains the full R-orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, \mathsf{R}P_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

Theorem 6.10 (Witness enclosure)**.** *For every exact-one-supercritical parameter pair (a, b) constructed above,*

$$\Lambda(K_{\mathrm{wit}}(a, b)) \geq 1.$$

Theorem 6.11 (Zero-gap obstruction)**.** *There is no cover of H by open unit equilateral triangles $U_C, U_0, \ldots, U_5$ such that every U_i is* Vd0*, the six vertex triangles cover ∂H, and exactly one V triangle is supercritical.*

Proposition 6.12 (Cases excluded by the nine-point obstruction)**.** *Theorem 6.11 excludes*

(1) *the* CE0*, $N_+ = 1$, all-*Vd0 *case;*
(2) *the no-gap* CE1 *and* CE2*, $N_+ = 1$, all-*Vd0 *cases.*

7. Exhaustive Assembly Statement

Table 1 gives the exhaustive case organization for the main theorem. In the complete paper, every row is excluded by the trace-length, boundary-propagation, area-loss, or nine-point results recorded above. The resulting conclusions are Theorem 1.1 and Corollary 1.2.

Appendix A. Exact Unequal-Radius Support-Arc Intersections

The two mixed cap overlaps in Method 4 are certified in the complete paper and its authenticated exact certificate package. This appendix retains the definitions and formal statements; calculation details and provenance data are omitted. The complete certificate package remains available in the repository.

A.1. **Rational radial upper envelopes.** Retain the notation of Section 6.1. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$
$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{87}$$

Put

$$\kappa = 4s^3 - 2s + m > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \ge 0). \tag{88}$$

Lemma A.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Reflect so that $z = a - b \ge 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$
$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$\bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}. \tag{89}$$

A.2. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (89), and put

$$\bar{\eta} = h(1 - \bar{c}), \qquad e = 1 - \bar{c}, \qquad t = 2\bar{c} - 1.$$

Lemma A.1 gives $0 < \bar{\eta} \le \eta = h(1 - c_*)$. The certificate targets the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = \mathsf{R}^{-1}C$. Define

$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C' \rangle,$$
$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$h^2 \Delta_0^2 = N_A N_C - \nu^2. \tag{90}$$

Define

$$Q_A = \Delta_0^2 - \|tA - eC'\|^2, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{91}$$

Lemma A.2 (Residual signs imply the mixed cap intersections)**.** *If* $Q_A, Q_C \geq 0$, *then*

$$I(C, 2\bar{\eta}) \cap I(C + \mathsf{R}A, \bar{\eta}) \neq \varnothing,$$

and

$$I(C + \mathsf{R}A, \bar{\eta}) \cap I(\mathsf{R}A, 2\bar{\eta}) \neq \varnothing.$$

Consequently the same intersections hold with η *in place of* $\bar{\eta}$*.*

A.3. **Exact residual statement.** For $B \in \{L, T\}$ and $X \in \{A, C\}$, the exact certificate uses eight core polynomials in $\mathbb{Z}[U, z]$, specified by

$$\operatorname{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\}. \tag{92}$$

Theorem A.3 (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with* $c_* > 2/3$*, all eight polynomials in* (92) *are nonnegative. Hence* $Q_A, Q_C \geq 0$*, and the two mixed cap intersections required by Proposition 6.8 hold.*

Remark A.4 (Certificate availability). The authenticated exact data and verifiers remain available in the repository package supporting the complete paper.